

Coming to remember

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Abstract. This paper is concerned with change-of-state uses of memory predicates like English *remember*. Change-of-state ‘remember’ appears to denote a change from not remembering to (stative) remembering – roughly ‘(slowly/suddenly) coming to remember’. It is, therefore, suitable to indicate constructive episodic memory retrieval. I argue that these uses are best captured as regular stative remembering, but with only partial – i.e. sub-atomically quantified – events as their object. I predict that remembering some, but not all, details of an event shows homogeneity effects arising from non-maximality. Speakers’ extra-linguistic projection of motion and change onto the third truth-value leads to the common eventive interpretation. No ontologically meaningful notion of change is required in the semantic composition or lexicon.

Keywords: change-of-state attitudes · memory predicates · subatomic event quantification · non-maximality · homogeneity effects.

1 Introduction

The semantic investigation of memory predicates like English *remember* has long been limited to cases like (1) where an agent x stands in the remember-relation to a simple *that*-proposition p – and x ’s standing in this relation to p is understood as a stable property of x or a state.

- (1) Mia remembers exactly that it was her brother’s birthday yesterday (– even if she might not be thinking about it right now).

On a standard Hintikka (1969) analysis, (1) is true if and only if all of Mia’s ‘(exact) memory worlds’ (p. 91) are worlds in which ‘*that it was Mia’s brother’s birthday yesterday*’ is true. (See Higginbotham, 2003, for a more recent account in this general spirit.) The lexical core of predicates like *remember* under such analyses does not come with any temporal or aspectual demands on the attitude-holding. This matches the usage of (1): The acceptability of the bracketed continuation shows that conscious mental entertainment of p at the reference time is not a part of the truth-conditions of (1). However, there are cases like (2-c) and (2-d) where the remembering seems to be an event of some sort or is only being realized at the reference time. Furthermore, other objects apart from simple *that*-

propositions can be remembered, see (2-b) and (2-d).¹ Adding the repetition of (1) as (2-a), the pattern in (2) emerges: Event-likeness of the object of remembering and of the realization of the attitude of remembering are orthogonal and all four possible combinations are attested. I refer to the case where the attitude appears (pre-theoretically) to be an event as ‘eventive remembering’ (roughly: remembering *as* an event; (2-c), (2-d)). I refer to the case where the object of remembering appears to be an event (again, pre-theoretically) as ‘experiential remembering’ (roughly: ‘remembering *of* an experienced event’; (2-b), (2-d)).²

- (2) a. Mia remembers exactly **that** it was her brother’s birthday yesterday
(– even if **she might not be thinking about it** right now). **stative, fact-only**
- b. Mia remembers exactly **how** she danced on the table yesterday (–
even if **she might not be thinking about it** right now). **stative, experiential**
- c. ... **and then, suddenly,** Mia remembered **that** it was her brother’s
birthday yesterday. **eventive, fact-only**
- d. ... **and then, suddenly,** Mia remembered **how** she danced on the
table yesterday. **eventive, experiential**

In recent years, the linguistic focus has shifted away from the exclusive analysis of the stative+fact-only case that is exemplified by (2-a). However, existing accounts still commonly focus on only one or two of the four attested cases in (2). For instance, Özyıldız (2021) – the first broad investigation of the event properties of attitudes, to my best knowledge – discusses the stative / eventive (‘dynamic’) axis in some depth, but disregards experiential complements. Liefke (2024) implicitly only considers eventive cases (as pointed out by Smith 202x); Rosina and Liefke (2025b) explicitly try to account only for the stative ones. Stephenson (2010)’s notion of remembering as a ‘vivid attitude’ pools experiential and eventive together, contrasting it only with the stative+fact-only case which shares none of the two properties with ‘vivid attitudes’. Smith (202x) focuses on the experiential+stative case in Kurdish; and, finally, in Rosina (202x) I discuss the stative/eventive distinction – but by backgrounding the fact-only/experiential distinction.

¹ – but also cases like ‘remember the party’, ‘remember to water the plants’, ‘remember how to play the piano’, or ‘remember whether p’, which are not discussed in this paper.

² The experiential cases (2-b) and (2-d) are more commonly discussed in the form of *remember*+GERUNDIVE complement clauses (‘remember dancing’; see e.g. Stephenson 2010; Bernecker 2010). For the purpose of this paper, I assume that the semantics of such sentences is relevantly similar to ‘remember how p’ under a non-manner *how* (Umbach et al., 2022) reading. This is because the latter avoids discussion of the PRO subject in ‘remember swimming’ and because there is a fully compositional account of a German equivalent of ‘remember how p’ provided in Rosina and Liefke (2025b). An analog semantic distribution of German memory predicates and English *remember* is supported experimentally in Rosina and Liefke (2025a).

Unsurprisingly, in the light of the multiplicity of uses of memory predicates, some previous accounts have postulated ambiguity (Stephenson, 2010) or polysemy (Liefke, 2023) of memory predicates along one of the axes. The aim of this paper is to complement the analyses in Rosina and Liefke (2025b) and Rosina (202x) in such a way that a coherent story of the *interplay* of experientiality and eventivity of the attitude of remembering emerges. The resulting full account makes sense of the whole pattern in (2) while requiring only one, non-polysemous, lexical entry of *remember* (and, potentially, of its cognates in other languages). This paper fills a gap in the argument for such a full, unified account of *remember* in (2) by providing a semantics of the most complicated case, a ‘coming to remember’-reading of the eventive+experiential case, (2-d).

On this reading, (2-d) introduces an event in which Mia suddenly (/slowly /effortfully /...) comes to (statively) remember the dancing event. It seems that remembering is both the endpoint of the change and the change itself in this case – an observation which strongly suggests two separate lexical entries for stative and ‘change of state’-remembering at first glance. Foreshadowing, I instead analyze the ‘coming to remember’-reading of (2-d) such that a context is accommodated in which Mia is remembering some, but not all, details of the dancing event.³ Mental change is merely extra-linguistically projected onto this state of affairs by the hearer to make the state of (half-)remembering compatible with the event-demanding environment in (2-d).

I combine this idea with existing analyses of the *how*-complement in (2-d) as an informationally rich proposition (Umbach et al., 2022; Liefke, 2023; Rosina and Liefke, 2025b), and of remembering as non-novel evidential knowledge (Rosina, 202x). The resulting full semantic account of ‘*x remember how/that p*’ in all of the sentences in (2) contains no events in the semantic composition whatsoever – neither in the denotation of the complement nor via a lexical property of the attitude. Instead, all cases in (2) turn out stative (in a sense) and fact-only (in a sense). Experiential remembering, (2-b) and (2-d), is reduced to remembering of informationally rich propositions (Rosina and Liefke, 2025b). Eventive remembering, (2-c) and (2-d), is reduced to a temporally bounded and/or non-maximal state (Rosina, 202x, and this paper). A salient interpretation of this move is a conservative defense of stative+propositional Hintikka (1969)-style approaches to the attitude of remembering – but adapted to fit more nuanced observations concerning event-likeness, and at the cost of uncommon claims about the interfaces of semantics, pragmatics, and extra-linguistic cognition.

This paper is structured as follows: Section 2 introduces the puzzle of eventive+experiential uses of *remember* in some more detail and delineates ‘change of state’-uses as a sub-case of these. Section 3 develops my semantic proposal for these uses, based on a plural propositional analysis of events. Section 4 concludes and addresses some bigger-picture questions.

³ This is roughly parallel to Özyıldız (2021)’s treatment of remembering partial answers to wh-questions, see the outlook in Section 4

2 An event of remembering an event?

Consider again (2-d), repeated as (3), which I have characterized as eventive + experiential.

- (3) ... **and then, suddenly,** Mia remembered **how** she danced on the table yesterday. **eventive, experiential**

For the upcoming initial discussion, I will assume, simplifying Rosina and Liefke (2025b), that experientiality can be reduced to informational richness of the embedded proposition. In effect, the idea there is that *how* in ‘*how p*’ identifies the salient situation which makes *p* true and restricts *p* by other propositions which were true in this situation (see Rosina and Liefke, 2025b, for the composition). Let’s take a context in which the salient *p*-situation (that is, the salient ‘Mia table-dancing yesterday’-situation) makes *q* (‘*that it was dark outside*’), *r* (‘*that Abba was playing*’), and *s* (‘*that there were flashing disco lights*’) also true, amongst other propositions. In this context, *remembering how p* requires remembering of $p \wedge q \wedge r \wedge s$ (possibly intersected with further circumstantial propositions). Consequently, (3) is equivalent to (4) when evaluated against the given context.

- (4) ... **and then, suddenly,** Mia remembered **that** she danced on the table yesterday and it was dark outside, and Abba was playing, and there were flashing disco lights, and ... **eventive, propositional (inform. rich)**

In as far as ‘*remembering how p*’ is an episode towards an event at all, then, this event is modelled propositionally. No event-type is required in the composition. In this paper, I suggest that the propositions *p* – *s* in the above sketched context are suitable to model mereological *parts* of such pre-theoretic events (instead of ‘manners’, as in Rosina and Liefke, 2025b), see Section 3.

With this reduction of experientiality to informational richness in place, the only thing that stands in the way of a uniform treatment of *remember* in the sentences in (2) in Section 1 is eventivity. Recall that ‘eventivity’ refers to the (seeming, pre-theoretic) event-likeness of (the realization of) the attitude itself. Intuitively, what makes this reading the predominant one in (3)/(4) is *suddenly*, since a state cannot be sudden (in a direct modification sense), but its ‘coming about’ can be.

Note that, at closer inspection, (3)/(4) has two distinct readings, both of which are eventive in the following broad sense: They necessarily entail that a (pre-theoretic) event existed in which Mia ‘got into’ the state of remembering *how p*.⁴ However, the nature of this entailment varies between the two readings. To disentangle the two, consider the contrasts and parallels in (5).

- (5) a. ... and then, suddenly, Mia remembered how *p*.
b. ... and then, suddenly, the screen was black.

⁴ This is also the sense of ‘eventive’ used in Rosina (202x), where no sub-readings are disentangled, cf. Section 3.

- c. ... and then, suddenly, the screen turned black.
- d. ... and then, slowly, Mia remembered how *p*.
- e. ??... and then, slowly, the screen was black.
- f. ... and then, slowly, the screen turned black.
- g. ??... and then, slowly, Mia knew how *p*.

I suggest that *remember* in (5-a) has a reading which is analogous to (5-b) in the relevant respects and one that is analogous to (5-c) in these respects. *suddenly* in (5-b) and under the first reading of (5-a) modifies the reference situation high up in the syntactic structure, leading roughly to the meaning paraphrased in (6).

- (6) ... and then, the situation had suddenly evolved such that, as a result, Mia was in the state of remembering how *p*.

I call this reading a ‘temporary state’ reading, in line with Özyıldız (2021), p.172. While it, again, requires some event in a broad sense (as opposed to continuous states), remembering itself can straightforwardly remain a state and has no eventive lexical properties. (By analogy, ‘the screen being dark’ in (5-b) remains a state even if it comes about in an event due to *suddenly*.) The other reading of (5-a) is eventive in a way which is more challenging for the overall account proposed in this paper. This reading roughly corresponds to Özyıldız (2021)’s ‘dynamic’ remembering,⁵ and to Smith (202x)’s notion of a ‘change of memory state’. On this ‘change of state’-reading, it appears as if the change itself is denoted by *remember*. This is witnessed by the fact that stative remembering (at least of the full event) is not entailed by ‘change of state’-remembering – it may even be excluded. Take the sequence in (7):

- (7) Shhh, don’t disturb Mia! She is finally remembering how she lost her keys yesterday.

The speaker’s advice would make no sense if Mia already (statively) remembered the key-losing event (fully). The idea that the remembering itself is a process under this reading might be formalized by the use of degrees, potentially. It is in line with the fact that *slowly* can modify *remember* under this reading in (5-d), like it can modify ‘*turning black*’. States like ‘*the screen being black*’, (5-e), or knowing, (5-g), cannot be modified by *slowly* in this way.

This ‘real’ ‘change of state’-reading is the most genuinely eventive at first sight. There seems (for now, but see Section 3) to be no way around another lexical entry of *remember* under which the attitude comes in degrees. To sum up the delineation of broadly eventive readings of *remember*, genuine ‘change of state’-readings behave event-like both with respect to modification by *suddenly* and with respect to modification by *slowly*. In contrast, ‘temporary state’-readings

⁵ – which is, however, further characterized as an accomplishment (p. 150, 172, 176), while I discuss it as a possible achievement in (8-c). The important property for the current purpose seems to be telicity – a property shared by accomplishments and achievements, see Özyıldız (2021), p. 12, and references there.

pass some (e.g. modification by *suddenly*), but not all (e.g. not modification by *slowly*), eventivity tests. This is in line with Sorace (2000)’s gradient notion of the scale from state- to process-likeness (proposed in the context of auxiliary selection).

I will now focus on explaining the harder case of ‘change of state’-uses for most of the rest of the paper. A first attempt at a paraphrase of (8-a) is given in (8-b). The corresponding attempt at a lexical entry for ‘change of state’-remember is provided in (8-c).⁶

- (8) a. ... and then, slowly, Mia remembered how she danced on the table yesterday.
 b. ... and then, Mia’s state was slowly changing from one where she did not remember how she danced [...] to one where she did remember how she danced [...] (rejected later)
 c. $\text{do}'(x, [\text{retrieve.episodic.memory}'(x)])$
 $\wedge \text{PROC constructing.mental.scenario}'(x)$
 $\wedge \text{INGR remember.state}'(x, \text{how } p)$ (rejected later)

One obvious disadvantage of this solution is the necessary ambiguity or polysemy of *remember* between a ‘change of state’ and a stative lexical entry. Furthermore, none of the entries is built ‘on top’ of the other in an additive, intersective way (where, e.g., the process would be the state *plus some other demands*). Rather, attribution of the attitude denoted by one polyseme would entail that attribution of the other fails. (If you are in the course of remembering, you don’t fully remember, and if you do fully remember, you are not in the course of remembering.) This might, of course, not be unpalatable for some scholars, depending on varied background beliefs. However, there is yet another problem with this ambiguity/polysemy: There seem to be no other cases of achievements where the end point of the achievement is lexicalized in the same way as the achievement (while both have the same object). The two polysemes would relate to one another like ‘*come to know*’ or *learn to know*; like ‘*going home*’ to ‘*being home*’; like ‘*the screen turning black*’ to ‘*the screen being black*’ etc.

⁶ The ‘activity’ part of the lexical composition (i.e. the first conjunct in (8-c)) is a standard part of the lexical semantics of achievements and is motivated by modification by ‘*with a great deal of effort*’ in (i-a). In contrast, someone else’s effort must be made salient for (i-b) to be acceptable:

- (i) a. ... and then, with a great deal of effort, Mia remembered how *p*.
 b. ?... and then, with a great deal of effort, the screen {was/turned} black.

English *remember* might also have a reading under which the process is happening involuntarily, such that the process towards the endpoint of stative remembering is attributed solely to circumstantial triggers (‘*couldn’t help but remember*’). German *einfallen* (PREFIX-fall) is a memory-related verb which assigns the rememberer a recipient rather than an agent role. I must leave the investigation of the activity/agent component of memory predicates for future research.

In the upcoming Section 3, I argue that we do not have to decide between an ambiguity/polysemy approach to the eventive/stative distinction and failing to capture the event-like properties of ‘change of state’-*remember*. Instead, we can have the cake and eat it with a few non-standard assumptions about natural language ontology and extra-linguistic cognition.

Before I move on, I would like to comment on an alternative to polysemy and ambiguity proposed by Özyıldız (2021): After diagnosing that ‘remember’ seems to alternate freely between giving rise to stative and dynamic descriptions’ (p.150), Özyıldız (2021) suggests that *remember* is underspecified in this respect:

‘We cannot always meaningfully talk about whether a verb has such and such lexical aspectual property – as these properties are sometimes determined by a verb taken together with its arguments and modifiers (atelic ‘run’ or ‘drink booch’ vs. telic ‘run to the park’ or ‘drink the booch’).’ (p.150)

While this makes sense to explain the cases discussed by Özyıldız (2021), it does not suffice to explain a difference between continuously stative, temporally stative and ‘change of state’-readings of one and the same sentence ‘*Mia remembered how p*’, since these logically distinct sentences share the same arguments (at least at the surface level). Therefore, I remain committed to finding one core lexical meaning of *remember*, if possible, and derive the other readings indirectly from that entry in interaction with other factors. With this clarification in place, I now move on to introduce my positive proposal. This proposal shares with Özyıldız (2021) the general idea that the complement clause of *remember* plays a central role in determining the event-like properties of the full sentence. However, my proposal centers the experiential case – where a pre-theoretic event is remembered – and proposes a different kind of interaction between the attitude and its object.

3 Proposal: Knowing event parts

In this section, I explore the idea that ‘change of state’-uses of *remember* come about by the composition of a regular stative lexical entry of *remember* with a non-maximal part of the complement event. The two main ingredients of this analysis are the following: (i) Stative remembering is decomposed further into RE-’ (morphologically transparent restitution) and KNOW’ (denoted by *remember*). Thus, the attitude of remembering consists in the restitution of the state of knowing (see Rosina, 202x). And (ii), mereological parts of (pre-theoretic) events can be modelled as propositions which are true at the intuitive event. The idea in (ii) is applied to an experiential and to an eventive attitude for the first time here. The more general proposal of modelling attitude content as mereologically understood propositions can be found in Haslinger and Schmitt (2021), who apply it to ‘*believe that*’. The fact that plural propositional complements enter cumulative relations (see *ibid.*) supports this conception of attitude content independently. I briefly present (i), based on Rosina (202x), in Section 3.1; followed by a discussion of (ii) and a sketch of the suggested composition in Section 3.2.

3.1 Remembering as restituted knowledge

In Rosina (202x), going back to Rosina and Liefke (2025b), the state of remembering is characterized as a kind of knowledge in a morphologically decompositional way. This general idea (without the compositionality) is prominent in epistemic philosophical theories of remembering like Hörl (2022) a.o. More specifically, knowledge is characterized via evidential support (indicated by *evi* in (9)) for true propositions in Rosina and Liefke (2025b). Remembering – understood as the cross-linguistic semantic commonality between memory predicates – presupposes this evidence-based knowledge about a previous time and asserts it about the reference time. This previous time may precede the reference time directly or indirectly – in other words, remembering is non-novel knowledge (Rosina, 202x, (9)). Cross-predicate variation concerning the availability of different stative and eventive uses is attributed to differences between ‘*still know*’, ‘*know again*’ and ‘*re-know*’ which come with different temporal requirements on the presupposed knowledge.

- (9) **REMEMBER**’ =ex. (9) in Rosina (202x)

$$\llbracket \text{non-novelly know} \rrbracket^{\text{a},c} = \lambda p : p_{\text{a}} . \lambda x : \frac{(\exists t)[(t \prec t_{\text{a}}) \wedge \text{evi}(x, p)_{\langle w_{\text{a}}, t \rangle}]}{\text{evi}(x, p)_{\langle w_{\text{a}}, t_{\text{a}} \rangle}}$$

 presupposed: $p=1$ in @; at some time (directly or not) preceding t_{a} ,
 $\text{evi}(x, p)$.
 asserted: $\text{evi}(x, p)=1$ in @ currently

Zooming in on the concrete English verb *remember*, its ‘*re-know*’-type semantics has two important effects for event-like properties of *remember* which are not pursued in Rosina (202x): temporal boundaries and a stative lexical core. These are discussed in turn below.

Temporal boundaries. The temporal gap in the state of knowledge (effected by *re-*, as opposed to *still*) introduces a temporal boundary. This temporal boundary has a similar effect to *suddenly*, discussed in Section 2: It makes eventive readings in a broad sense – ‘temporary state’ readings – available. This is illustrated by the analysis of *remember* in sentence (10-a) on its ‘temporary state’ reading in (10-b), (10-c).

- (10) a. (... and then, suddenly,) **Mia remembered how *p***. =(5-a)
 b. (... and then, the situation had suddenly evolved such that, as a result,) **Mia was back in the state of knowing how *p***.
modified (6)
 c. $\llbracket \text{Mia remember how } p \rrbracket^{\text{a},c} = 1$ iff

$$\frac{\text{how } p_{\text{a}} . (\exists t)[((t \prec t_{\text{a}}) \wedge (t \not\prec! t_{\text{a}})) \wedge \text{evi}(\text{Mia}, \text{how } p)_{\langle w_{\text{a}}, t \rangle}]}{\text{evi}(\text{Mia}, \text{how } p)_{\langle w_{\text{a}}, t_{\text{a}} \rangle}}$$

 presupposed: $\text{how } p=1$ in @; at some time preceding – but not directly – t_{a} , $\text{evi}(\text{Mia}, \text{how } p)$.
 asserted: $\text{evi}(\text{Mia}, \text{how } p)=1$ in @ currently

I weakly suggest that no event is needed in the corresponding composition at all for *suddenly* to modify in some sense the coming about of the reference situation high up in the structure. (See Section 2 for discussion of *suddenly* and Section 3.2 for an extra-linguistic account of change.) But if someone prefers to understand the meaning of (10-a), (10-b) including the bracketed parts via such an event, nothing in my general account speaks against this. *remember* still stays stative in the weak sense of a temporary state in this case, because the event is elsewhere in the structure.

The analysis in (10) does not straightforwardly extend to ‘change of state’-readings which I have identified by modification by *slowly* in Section 2. These require more than just a temporal bound. Thus, the availability of ‘*slowly remember how p*’ remains to be explained. (This is done in Section 3.2.)

A stative-only lexical core. The morphological boundness of *re-* makes it such that its *again*-like meaning can only be applied to the state of knowing directly (as illustrated in (10)), not to an event in which this knowledge comes about⁷ – even if there was such an event higher up in the structure. As a consequence, the morpho-semantics of *remember* according to Rosina (202x) not only does *not explain* ‘change of state’-readings – it blocks them. (Since *re-* would intervene between the state of knowledge and its suggested coming-about event.) If one were to maintain an ambiguity/polysemy approach as sketched in Section 2, the ‘change of state’-polysemy could not be related to its morphological make-up at all. The morpheme *re-* would play no role whatsoever, and knowledge only as a not-yet-achieved goal with an unclear compositional role.

In what follows, I suggest instead that *remember* only ever has the ‘temporary state’-meaning in (10): Non-temporary, maximally continuous state readings are excluded by the temporal boundary introduced by *re-*. (But these boundaries can be very far from the reference time, so there might not be much practical effect of this.) The seeming ‘change of state’-use of *remember* can be explained as a temporary state as well – just as one towards a non-maximal object. I get to this case now.

3.2 Partial events as plural propositions

After a long focus on different forms of eventivity (the event-like properties of the attitude), let us focus on the informationally rich proposition *how p* (effecting event-like properties of the object). In Section 2, I have introduced *how* as causing an intersection of *p* with further properties which were true at the salient *p*-event. Sentence (11-a) is equivalent to (11-b) when evaluated against a suitable context.

- (11) a. ... and then, suddenly, Mia remembered how she danced on the table yesterday.

⁷ The latter would lead to a repetitive reading of *again*, where the recall-event is repeated, see Rosina 202x, based on Lechner et al. (2015); Iyer (2022).

- b. ... and then, suddenly, Mia remembered that she danced on the table yesterday and it was dark outside, and Abba was playing, and there were flashing disco lights, and ...

Deviating from Rosina and Liefke (2025b) and Umbach et al. (2022), I propose that the resulting informationally rich complement clause in (11-b) is best understood as a plural proposition $p + q + r + s$ (instead of an intersection or the manner of an event token).⁸ A maximally informationally rich proposition is a full specification of the corresponding (pre-theoretic) event. It is also the maximal theoretically possible salient plurality (but see the discussion below).

Propositions like ‘*Mia was wearing a red sweater and blue jeans*’ are parts of this maximal proposition (if they were true at the dancing-event). At the same time, they have parts themselves, e.g. the proposition ‘*Mia was wearing jeans*’. Such, syntactically speaking, seemingly atomic propositions can be said to have parts as well – for instance, ‘*Someone was wearing something*’ or ‘*Mia was wearing jeans or a skirt*’. All of these part-whole-relations are part of a mereological semi-lattice the linguistic application of which goes back to Link (1983).

Plural propositions like $p+q$ may be less intuitive ontological objects than plural individuals like $Mia+Ali$. Nevertheless, assuming their existence in language leads to a lot of correct predictions. For one, concrete situations hardly ever make the maximally possible plurality salient. Just like in other domains, plurality is subject to contextual domain restriction. Consequently, knowing the parts of an event, i.e. its details in a propositional form, never literally demands knowledge of *all* parts of an unrestricted domain of event parts. By analogy, *all boys* hardly ever refers to the maximal plurality $a+b+c+d+\dots$ of boys in the entire world.

The above analogy already suggests that ‘*know how p*’ is comparable to ‘*feed the cats*’ on my account: It needs to be contextually specified by the extension of ‘*how p*’ in the utterance world – e.g. $p+q+r+t$, corresponding to what was true at the salient p -event. In a similar way, *the cats* needs to be resolved to, say, $JURI+RIO$ in the utterance world. The literal interpretation of ‘*feed the cats*’ is ‘*feed all (salient) cats*’ (see e.g. Križ and Spector, 2021; Haslinger, 2022). Similarly, I suggest that the literal interpretation of ‘*know how p*’ is ‘*know all (salient) parts of the p-event*’, i.e. ‘*know the maximal plurality of (salient) propositions true at the p-event*’. Only this complete knowledge makes the sentence truth-conditionally true. At the same time, the pragmatic acceptability threshold is less strict – cf. Haslinger (2022) a.o. for the corresponding plural individual case. Less than perfect knowledge of the salient details of an event sometimes suffices to license the utterance of sentences like ‘*Mia remembers how p*’. I suggest to identify such cases with the seeming ‘change of state’ readings. To support this idea, I will first show that ‘change of state’-readings of *remember*-reports show many of the same semantic effects as (other) relations between an agent and a

⁸ See Haslinger and Schmitt (2021) for a predecessor of the general spirit of using plural propositional complements to solve attitudinal puzzles.

non-maximal plural object. Then, I will make a suggestion as to where the clear intuition of change comes from, if it is not lexically encoded.

Note that a context can be constructed in which ‘*know how p*’ under a ‘change of state’-reading shows homogeneity effects similar to the ones observed with definite plurals (for the latter, see Križ 2019):

- (12) a. CONTEXT: Mia is trying very hard to mentally reconstruct her whole previous evening in order to find out how she has lost her keys. She is already able to recall many details which she had forgotten and which indicate that she should look for her keys in the discotheque ‘Mamma Mia’.
- b. ?Mia remembers how she lost her keys.
- c. ?Mia doesn’t remember how she lost her keys.
- (13) a. CONTEXT: 7 of 10 windows are open.
- b. ?The windows are open.
- c. ?The windows are not open.
- (14) a. A: ‘Does Mia remember how she lost her keys yesterday?’
- b. B: **Well**, she is slowly remembering it right now.
- c. B: **Well**, not well enough yet to find them.
- d. B: **Well**, well enough at least to find them.
- (15) a. A: ‘Are the windows closed?’
- b. B: **Well**, most of them are.
- c. B: **Well**, not all of them are.
- d. B: **Well**, enough of them for it to not get too cold.

The pragmatic acceptability threshold of required remembered details is context-dependent and often vague (Haslinger, 2022). Consequently, there are contexts in which it is acceptable to utter (12-b) even if Mia remembers only some, but not all, relevant details of the key-losing event. At the same time, the semantic truth-value remains neither True nor False regardless of the utterance context. Instead, the sentence has a third truth value, #. This explains the observed homogeneity effects. What I am claiming, in effect, is the following: Seeming ‘change of state’-*remember* is just regular stative remembering in a context in which the memory report has the truth value #, but can nonetheless be uttered due to pragmatic acceptability.

An eventive environment – as created for instance by modification by *slowly* – requires *something* to change, at *some* level. This is not compatible with stative remembering of a whole event (i.e. of all of its propositional parts). I suggest that this is the reason why knowledge of only a non-maximal event part is assumed by the hearer in such cases – in order to be able to process the sentence at all. The impression of change arises merely as an extra-linguistic assumption; similar to the way in which the utterance ‘I have closed half of the windows already’ evokes a scenario in which the cumulative state of the windows changes from (all) open to (all) closed.

A last question that has remained open is why the direction of change is so clearly indicated by ‘change of state’-*remember* if it denotes only remembering of partial events. In other words, why is ‘change of state’-*remember* not used to report (slow) forgetting? The answer to this question lies in the presupposed content of *remember* (see (10-c) in Section 3.1): The previous knowledge is presupposed of a *non*-preceding time. Therefore, the most recent starting point which is in the common ground is one of non-knowledge. Given that the common ground is now updated by ‘partial knowledge’, a change from ‘none’ via ‘some’ towards – projecting into the future – ‘all parts of the event’ being known to the agent is constructed. The parallel to plural DPs can still be upheld: ‘Half of the children are already awake again’ leads to the mental construction of a change from (all) sleeping children to (all) awake children. To sum up my analysis of ‘change of state’-*remember*, the assumed change is extra-linguistic, and all uses of *remember* can be unified under one - stative, propositional - lexical entry.

4 Conclusion

This paper brings the debate whether all attitudes are propositional together with the long-neglected discussion on the eventive properties of (the realization of) attitudes themselves (Özyıldız, 2021). It does so by studying the case of eventive+experiential uses of English *remember*. These uses seemed, at first sight, to report an event of remembering an event.

I argued that some eventive uses can be reduced to temporally bounded states straightforwardly. The remaining ‘change of state’-uses uniquely allow for modification with *slowly*. The existence of such uses gives rise to the puzzle that the process and the endpoint of the change are denoted by the same verb on the surface: Mia (eventively) remembering how p' would be equated with Mia getting into the state of (statively) remembering how p .

While this seems to suggest ambiguity or polysemy between stative and ‘change of state’-uses of *remember*, I proposed that we can do without it: Seeming ‘change of state’-*remember* is an interpretative strategy which resolves a semantic clash between the stative nature of (all occurrences of) *remember* on one hand and environments (like modification by *slowly*) which force a change to happen. This change is constructed extra-linguistically if needed. Its construction draws on the presupposed lost-knowledge startpoint of *remember* taken together with its assertion of current non-maximal remembering. This idea is supported by ‘change of state’-remembering showing homogeneity effects similar to the ones observed with plural DPs. The parallel is strengthened by treating sub-atomic event quantification on par with super-atomic propositional plurality.

This analysis of ‘change of state’-*remember* paves the way for my overarching claim – cumulatively supported by Rosina and Liefke (2025b); Rosina (202x) and this paper – that all uses of English *remember* are based on the same lexical entry. This entry is unambiguously specified (viz. Özyıldız, 2021) as stative and takes only propositional objects.

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